FDS LAB DAY 1 WORKOUT QUESTIONS

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1. Create a DataFrame Question: Create a DataFrame from the following dictionary: python CopyEdit data = { 'Name': ['Alice', 'Bob', 'Charlie', 'David'], 'Age': [25, 30, 35, 40], 'City': ['New York', 'Los Angeles', 'Chicago', 'Houston'] } Display the DataFrame.

```
[2]
import pandas as pd

[3]
data = {
    'Name' : ['Alice', 'Bob', 'Charlie', 'David'],
    'Age' : [25, 30, 35, 40],
    'City' : ['New York', 'Los Angeles', 'Chicago', 'Houston'],
}
df = pd.DataFrame(data)

[4]
df
```

	Name	Age	City
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Charlie	35	Chicago
3	David	40	Houston

Next steps: Generate code with df | New interactive sheet

2. Filtering Rows Question: From the above DataFrame, filter out the rows where the age is greater than 30

```
[5]
✓ 0s
df[df['Age']>=30]
```

	Name	Age	City
1	Bob	30	Los Angeles
2	Charlie	35	Chicago
3	David	40	Houston

3. 3.Adding a New Column Question: Add a new column Salary with values [50000, 60000, 70000, 80000] to the existing DataFrame

```
df['Salary'] = [50000,60000,70000,80000]

df
```

	Name	Age	City	Salary
0	Alice	25	New York	50000
1	Bob	30	Los Angeles	60000
2	Charlie	35	Chicago	70000
3	David	40	Houston	80000

4. 4. Update Column Values Question: Increase the salary of all people by 10%.

```
df['Salary'] = df['Salary'] * 1.10

df
```

	Name	Year	City	Salary
0	Alice	25	New York	55000.0
1	Bob	30	Los Angeles	66000.0
2	Charlie	35	Chicago	77000.0
3	David	40	Houston	88000.0

5. 5. Drop a Column Question: Remove the City column from the DataFrame

```
df.drop("City",axis=1)
```

	Name	Age	Salary
0	Alice	25	50000
1	Bob	30	60000
2	Charlie	35	70000
3	David	40	80000

6. 6. Rename Columns Question: Rename the column Age to Years.

```
df.rename(columns = {"Age":"Year"},inplace=True)
```

```
df
```

	Name	Year	City	Salary
0	Alice	25	New York	50000
1	Bob	30	Los Angeles	60000
2	Charlie	35	Chicago	70000
3	David	40	Houston	80000

7. 7. Sorting Values Question: Sort the DataFrame by Salary in descending order

```
df.sort_values(by="Salary", ascending=False)
```

	Name	Year	City	Salary
3	David	40	Houston	88000.0
2	Charlie	35	Chicago	77000.0
1	Bob	30	Los Angeles	66000.0
0	Alice	25	New York	55000.0

8. 8. Grouping and Aggregation Question: Create a new DataFrame: python CopyEdit data = { 'Department': ['HR', 'IT', 'HR', 'IT'], 'Employee': ['Alice', 'Bob', 'Charlie', 'David'], 'Salary': [50000, 60000, 55000, 65000] } Group by department and find the average salary.

```python
import pandas as pd

data = {
    'Department': ['HR', 'IT', 'HR', 'IT'],
    'Employee': ['Alice', 'Bob', 'Charlie', 'David'],
    'Salary': [50000, 60000, 55000, 65000]
}

df = pd.DataFrame(data)

print(df)
```

```
  Department Employee  Salary
0         HR    Alice   50000
1         IT      Bob   60000
2         HR  Charlie   55000
3         IT    David   65000
```

9. 9. Merge Two DataFrames Question: Given two DataFrames: python CopyEdit df1 = pd.DataFrame({ 'ID': [1, 2, 3], 'Name': ['Alice', 'Bob', 'Charlie'] }) df2 = pd.DataFrame({ 'ID': [1, 2, 4], 'Score': [85, 90, 88] }) Merge them on the ID column.

```python
import pandas as pd

df1 = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie']
})

df2 = pd.DataFrame({
    'ID': [1, 2, 4],
    'Score': [85, 90, 88]
})

result = pd.merge(df1, df2, on='ID')

print(result)
```

```
   ID   Name  Score
0   1  Alice     85
1   2    Bob     90
```

10. 10. Handling Missing Values Question: Create a DataFrame with some missing values and fill the missing data with appropriate default values.

```
import pandas as pd

df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [20, None, 22, None],
    'Salary': [50000, 60000, None, 70000]
})

print("Before filling missing values:")
print(df)

df['Age'] = df['Age'].fillna(0)
df['Salary'] = df['Salary'].fillna(0)

print("\nAfter filling missing values:")
print(df)
```

```
Before filling missing values:
      Name   Age   Salary
0    Alice  20.0  50000.0
1      Bob   NaN  60000.0
2  Charlie  22.0      NaN
3    David   NaN  70000.0

After filling missing values:
      Name   Age   Salary
0    Alice  20.0  50000.0
1      Bob   0.0  60000.0
2  Charlie  22.0      0.0
3    David   0.0  70000.0
```